

Standards-Based Mathematics Curricula and Secondary Students' Performance on Standardized Achievement Tests
Author(s): Michael R. Harwell, Thomas R. Post, Yukiko Maeda, Jon D. Davis, Arnold L. Cutler, Edwin Andersen and Jeremy A. Kahan
Source: *Journal for Research in Mathematics Education*, Vol. 38, No. 1 (Jan., 2007), pp. 71-101
Published by: National Council of Teachers of Mathematics
Stable URL: http://www.jstor.org/stable/30034928
Accessed: 07/04/2014 15:38

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at
http://www.jstor.org/page/info/about/policies/terms.jsp

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.



National Council of Teachers of Mathematics is collaborating with JSTOR to digitize, preserve and extend access to *Journal for Research in Mathematics Education.*

http://www.jstor.org

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

Journal for Research in Mathematics Education
2007, Vol. 38, No. 1, 71–101

Standards-based Mathematics Curricula and Secondary Students' Performance on Standardized Achievement Tests

Michael R. Harwell, Thomas R. Post, and Yukiko Maeda
University of Minnesota

Jon D. Davis
Western Michigan University

Arnold L. Cutler and Edwin Andersen
University of Minnesota

Jeremy A. Kahan
Ida Crown Jewish Academy, Chicago

The current study examined the mathematical achievement of high school students enrolled for 3 years in one of three NSF funded *Standards*-based curricula (IMP, CMIC, MMOW). The focus was on traditional topics in mathematics as measured by subtests of a standardized achievement test and a criterion-referenced test of mathematics achievement. Students generally scored at or above the national mean on the achievement subtests. Hierarchical linear modeling results showed that prior mathematics knowledge was a significant but modest predictor of achievement, student SES had a moderate effect, and increasing concentrations of African American students in a classroom were associated with a stronger effect of attendance on achievement. No differences on the standardized achievement subtests emerged among the *Standards*-based curricula studied once background variables were taken into account. The two suburban districts providing data for the criterion-referenced test achieved well above the national norm.

Key words: Cross sectional studies, Integrated curriculum, Multivariate techniques, Program/project assessment, Secondary mathematics

The current study examined achievement patterns of high school students enrolled in *Standards*-based curricula, with a focus on traditional topics in mathematics as measured by two nationally normed achievement tests. *Standards*-based curricula are curricula that were funded from a solicitation of proposals through the National Science Foundation (NSF) in the early 1990s (RFP NSF 91-100). These curricula were designed to be aligned with the National Council of Teachers of Mathematics'

The research was supported by the National Science Foundation under Grant No. NSF 96-18741. Any opinions, findings, and conclusions expressed are those of the authors and do not necessarily reflect the views of the National Science Foundation.

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

(NCTM) *Curriculum and Evaluation Standards for School Mathematics* (1989). Five secondary level mathematics curricula were funded by NSF: Contemporary Mathematics in Context (CMIC or Core-Plus) (Coxford et al., 1998), Interactive Mathematics Program (IMP) (Fendel, Resnick, Alper, & Fraser, 1997), Math Connections: A Secondary Mathematics Core Curriculum (Connections) (Berlinghoff, Sloyer, & Hayden, 2000), Mathematics: Modeling Our World (MMOW or ARISE) (Garfunkel, Godbold, & Pollak, 1998), and SIMMS Integrated Mathematics: A Modeling Approach Using Technology (SIMMS) (Lott & Burke, 1996). Schools represented in this study used CMIC, IMP, or MMOW.

This study builds on and extends our existing understanding of student achievement in *Standards*-based programs in three significant ways. First, we examine the impact of published editions of these curricula on student understanding as contrasted to earlier studies using field-test versions. All students in our study had participated in three levels of one of three *Standards*-based curricula. A high school level was defined to be the content equivalent of a yearlong course taught 1 hour per day (commonly referred to as a Carnegie unit). The three curricula (IMP, CMIC, MMOW) share major similarities including the recurring integration of topics within grade levels and a decreased emphasis on procedures and algorithms. They also differ from each other in various ways; the length of the units, the reality of the contexts, and the emphasized content are a few examples.

Second, this research concerns the use of *Standards*-based curricula as part of district-wide curricula adoptions. By examining the achievement of students in district-wide adoptions, we gain insights into more realistic situations where teachers are required, and have not necessarily volunteered, to teach *Standards*-based curricula. This provides a more accurate overall picture of expected student achievement, since many earlier field-test teachers were volunteers and therefore cannot be considered typical of all teachers who will eventually be expected to implement a *Standards*-based curriculum. Third, in studying student achievement, we employ hierarchical linear modeling (HLM) to account for the variability between classrooms and the interdependency of students within the same classroom (Kilpatrick, 2003). HLM allows us to consider student and classroom results simultaneously and to examine the role of classroom-level variables such as a teacher's hours of professional development and the *Standards*-based curriculum that a student experienced with various background variables taken into account.

Kilpatrick (2003) recommended that the next generation of curriculum research should focus on comparing different *Standards*-based curricula against one another on dependent variables measuring common goals. Following this recommendation, this study sought to deepen our understanding of the achievement of secondary students in *Standards*-based programs vis-à-vis multiple standardized-test measures.

THEORY UNDERLYING STANDARDS-BASED CURRICULA

In 1978, the National Council of Supervisors of Mathematics (NCSM), under the leadership of Ross Taylor, produced a document that redefined the "basic skills of

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

school mathematics." This document departed from current thinking at the time in that it identified the basics in broad strokes rather than specific and "behavioral" objectives that focused on skills that were computational in nature. Its fundamental concepts were incorporated into the NCTM's Agenda for Action (1980). As a result of these efforts, problem solving, spatial visualization and awareness, mathematics as communication, technological literacy, and appropriate computational skills became the new basics for school mathematics.

In 1989, the NCTM published *Curriculum and Evaluation Standards for School Mathematics;* in 2000, NCTM published a revised document, *Principles and Standards for School Mathematics*. This second document expanded upon and updated the 1989 *Curriculum and Evaluation Standards* and received wide and enthusiastic acceptance for reflecting best practices and current thinking in the field. The 1989 and 2000 *Standards* documents received wide endorsement for the integrity of the process used in their design and implementation and in the vision of school mathematics embodied therein. Among the many professional organizations endorsing these documents were the American Mathematical Society, Conference Board of the Mathematical Sciences, Mathematical Association of America, Mathematical Sciences Education Board, Society for Industrial and Applied Mathematics, American Statistical Association, Association for Women in Mathematics, and the Association of State Supervisors of Mathematics.

The 1989 and 2000 NCTM *Standards* documents were both broadly based on various cognitively oriented theories about the development of mathematics knowledge and the nature of student learning. Dienes suggested that mathematics knowledge should include the structural relationships within mathematical concepts, complemented by attention to their applications in the real world. The learning of mathematics involves the "apprehension of such relationships together with their symbolization, and the acquisition of the ability to apply the resulting concepts to real situations" (Dienes, 1969, p. 19). Related theories certainly include Piaget's theory of cognitive development about which much has been written (Furth, 1970; Ginsburg & Opper, 1988; Meece, 1997; Piaget, 1970). Vygotsky's sociocultural theory (Kozulin, 1986; Tudge & Winterhoff, 1993) and Dewey's positions (Dewey, 1916) emphasize that children are active learners, that they construct their own knowledge and that the social environment is a key setting for cognitive development and learning (Bruner, 1964; Bruner, Olver, & Greenfield, 1966). Lesh (1979) and Dienes (1969) extended these understandings by suggesting the importance of representation in the development and presentation of school mathematics curricula. One can see these and many other 'theoretical' positions reflected in the pedagogical guidelines as discussed within the process and curriculum standards.

Although the *Standards* documents are not completely research-based in a formal sense, they do reflect implications of the research conducted during the period preceding their publication. In 1989, the extent of the existing research was not adequate to completely inform a document with such a wide scope and extensive set of recommendations. The 2000 *Standards* document attempted to integrate existing research literature to a greater extent. Still, *Standards*-based curricula

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

cannot and do not claim to be completely informed by research, as they were developed to be consistent with several widely acclaimed documents (NCTM, 1989) that are only partially influenced by the results of empirical research. There is agreement, however, that these documents reflect best teaching practices and contemporary thinking about approaches to mathematics content development and the promotion of student learning.

STUDENT ACHIEVEMENT

Schoen and Hirsch (2003) found that CMIC students enrolled in Course 1 (9th grade) outperformed a group of similar students[1] studying prealgebra, algebra, and accelerated geometry on problems set within real-world contexts. Students in Course 2 (10th grade) outperformed comparable students enrolled in prealgebra and algebra on contextualized problems. There were no significant differences between Course 2 students and a matched group studying from accelerated advanced algebra. Huntley, Rasmussen, Villarubi, Sangtong, and Fey (2000) found that grade 11 students enrolled in CMIC performed significantly better than students enrolled in traditional curricula on problem-solving measures situated within a context.

Other studies (Cichon & Ellis, 2003; Lott et al., 2003) found that students in the *Standards*-based curricula Connections and SIMMS, respectively, outperformed students not in a *Standards*-based curriculum (i.e., a traditional curriculum) on contextual, open-ended problems. Webb (2003) found that students in the IMP curriculum performed significantly better than comparable groups of students learning from traditional mathematics programs on performance assessments involving statistics, probability, and combinatorics.

Standardized Achievement Tests

In order to satisfy concerns of parents and other stakeholders, evaluations of *Standards*-based curricula have also been conducted using standardized achievement tests. Schoen and Hirsch (2003) used the Iowa Test of Educational Development (ITED) Ability to do Quantitative Thinking, or ITED-Q (Feldt, Forsyth, Ansley, & Alnot, 1993) to assess student performance. CMIC Course 1 students performed $d = .17$ standard deviations higher on this test than a comparison group enrolled in prealgebra, and .57 standard deviations higher than a group enrolled in algebra, which were statistically significant differences. (Following Hedges & Olkin, 1985, p. 78, d is an effect size representing the difference in the means of two groups divided by the pooled within-groups standard deviation.) There were no significant differences between CMIC Course 2 students and those in algebra, geometry, or Accelerated Advanced Algebra. Webb (2003) found no significant differences between traditional and IMP students on the standardized test, Comprehensive Test of Basic Skills.

Abeille and Hurley's (2001) final evaluation report on the MMOW curriculum

[1] Students were matched based upon prior achievement, school, and gender.

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

describes student achievement data on two different standardized tests, the Stanford Achievement Test (SAT-9) and the ITED-Q. The authors found statistically significant increases in the number of MMOW students who scored in the top quartile of the ITED-Q when compared to students who had used traditional mathematics curricula at the same school in previous years. In another district, students learning from MMOW produced a mean score of 62 for the SAT-9 in 1999, expressed in normal curve equivalents (NCEs), and a mean of 60 in 2000. Both of these means were significantly above the national average of 50 for this test.

Symbolic Manipulation

Despite the successes of these curricula as described above, intense criticism from Web-based sources (e.g., Mathematically Correct—www.mathematicallycorrect.org) and others (Wu, 1997) has been leveled against *Standards*-based mathematics curricula. One of the areas of criticism has been whether *Standards*-based curricula provide students with sufficient practice to develop facility with symbolic manipulation tasks. This may be a valid criticism, for as Schoen, Fey, Hirsch, and Coxford (1999) point out, CMIC focuses on conceptual understanding with a decreased emphasis on symbolic manipulation.

There is mixed evidence with regard to this assertion that may be related to the use of technology in solving these types of problems. For instance, Huntley et al. (2000) found that 11th grade CMIC students did less well on symbolic manipulation tasks without the assistance of graphing calculators when compared to students enrolled in traditional mathematics programs, scoring on average .54 standard deviations lower. However, Schoen and Hirsch (2003) found no differences between students who had 4 years of CMIC and traditionally prepared students on a college mathematics placement exam emphasizing symbolic manipulation when graphing calculator technology (without symbolic manipulation capabilities) was available.

College Aptitude Tests

Lott et al. (2003) found no significant differences between SIMMS students and students in a traditional mathematics curriculum on the mathematics portion of the Preliminary Scholastic Aptitude Test (PSAT-M). Connections students scored just as well as traditional students on the PSAT-M, with prior achievement used as a covariate in the analysis (Cichon & Ellis, 2003). On the mathematics portion of the Scholastic Aptitude Test (SAT-M), there were no differences between Connections students and traditional students.

One of the strongest findings with regard to college aptitude tests is from Schoen and Hirsch (2003). They randomly assigned students to either CMIC or a traditional mathematics program in grade 9. Students' prior knowledge of mathematics at grade 9 in both groups was equivalent as measured by the Iowa Test of Basic Skills (ITBS). In grade 11, there were no statistically significant differences on the SAT-M between CMIC and traditional students. In another experimental study involving random assignment of students to classrooms, no significant differences were

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

detected between CMIC and traditional students on the American College Testing examination (ACT Math) in grade 11.

Webb (2003) found no significant differences on the SAT-M between 11th-grade students who had 3 years of IMP or a traditional mathematics program. One of the concerns of parents about *Standards*-based curricula is that higher achieving students will be disadvantaged by the program. Webb addressed this concern by examining the grade 11 SAT-M scores of IMP and traditional mathematics students who scored at or above the 76th percentile on the CTBS in grade 7 and found no significant differences between these two groups.

Overall, the data suggest that students studying from different *Standards*-based curricula do not perform differentially on the mathematics portions of the PSAT, SAT, or ACT college tests when compared to students learning from traditional curricula. This finding is relatively robust as it involves students from different geographical regions and in some cases involved random assignment of students to different curricula.

Ancillary Benefits of Standards-*based Curricula*

The focus of research regarding *Standards*-based mathematics programs has been on student achievement as summarized in the aforementioned studies. However, other benefits of studying in these programs have been documented. For instance, Webb (2003) found that students studying from IMP took more high school mathematics courses than students studying from traditional mathematics programs. Cichon and Ellis (2003) followed a group of 30 students in an inner-city school where 3 years of mathematics were required of all students. Although it was required that students study 3 years of mathematics, they found that 20 out of 30 students who had studied from Connections for 3 years elected to take mathematics as seniors. Moreover, 19 out of those 20 students chose to pursue postsecondary education. Typically, only 74% of students from this school continue on after high school.

The relationship between mathematics attitudes and achievement has been well researched (e.g., McLeod, 1994; Schoenfeld, 1989). Cichon and Ellis (2003) found that Connections students had significantly higher ratings in confidence and usefulness of mathematics on the Fennema-Sherman Mathematics Attitude Scale. Schoen and Pritchett (1998) found that students learning from CMIC Course 1 and 2 had more positive attitudes toward mathematics than a comparison group using a traditional mathematics program. For example, significant differences in favor of CMIC students were found on confidence in solving problems, believing that mathematical ideas make sense, and believing that mathematical ideas are inherently interesting.

Teacher Factors

There is much research indicating that teachers do not implement *Standards*-based curricula with fidelity (e.g., Lambdin & Preston, 1995; Truitt, 1998; Wilson & Lloyd, 1995). In order to better understand the relationships between implementation and student achievement, Schoen, Cebulla, Finn, and Fi (2003) used regres-

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

sion techniques to relate student achievement on the ITED-Q to particular teacher characteristics measured through teacher surveys and classroom observations. Nearly 1,500 students from 40 different teachers' classrooms participated in the Schoen et al study. The percentages of class time devoted to teacher presentation and whole-group discussion were both significantly negatively correlated with student achievement. Replacing activities or homework with skill supplementation and replacing CMIC (Coxford et al., 1998) tests with assessments that were less open-ended and more focused on symbolic manipulation were also significantly negatively correlated with student performance. On the other hand, summer workshop training, using individual interviews, the number of extending tasks assigned, the observer's rating, and collaboration between teachers were all positively significantly correlated with student achievement. These findings suggest that the commonly used teacher-dominated approaches to instruction, whose focus is on narrow and replicable routines, may not be the most efficient way to promote high levels of student achievement.

METHODOLOGY

Sampling

In the mid to late 1990s, the NSF, in an attempt to provide much needed professional development for school districts that adopted one or more of the new NSF funded *Standards*-based curricula, created the Local Systemic Change through Teacher Enhancement Initiatives (LSCs). The 47 funded LSCs (NSF 95-145) were "designed to engage entire school districts in the reform of science, mathematics and technology education, . . . to provide 47,000 teachers with professional development and . . . reach 1.6 million students in 240 school districts nationally" (Introduction, http://www.nsf.gov/pubs/1997/nsf97145/intro.htm).

The LSC discussed here, "Minneapolis and St Paul Merging to Achieve Standards Project" $(MASP)^2$, was one of these 47 Projects. $(MASP)^2$ provided professional development targeted to new NSF funded *Standards*-based curricula to over 1,100 middle grades and secondary teachers in 21 districts between 1997 and 2002. These teachers then provided *Standards*-based mathematics instruction to over 74,000 students in the 2000–2001 school year, and slightly larger numbers of students in the years since. Data from $(MASP)^2$ were used in this study. Although districts naturally varied by location (urban, suburban), especially striking differences emerged in enrollment and student characteristics, with mostly urban districts showing quite large enrollments and diverse student populations, and suburban districts tending to have smaller enrollments and relatively homogeneous student populations.

Following the distinction drawn by Cochran (1983), our target populations consisted of urban and suburban U.S. high school classrooms and students in a midwestern state who had completed at least three levels of high school mathematics in one of three *Standards*-based curricula. Our sampled populations, on the other hand, consisted of students and classrooms in seven purposively sampled urban and

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

suburban districts from those in $(MASP)^2$ who were subsequently invited to participate in the study of achievement reported here. Because of missing data in one district, six districts ultimately provided the high school data used in this study. These six districts were chosen for reasons of variation in location (urban and suburban), demographic variables, and school context. Mentors and project staff observed that there were high degrees of implementation and satisfaction with the curricula in some districts, whereas in others, the secondary mathematics faculty actively tried to undermine efforts to use *Standards*-based curricula.

The location, curriculum, rationale, assessment instrument, and enrollment for each of the sampled districts are shown in Table 1. One school district had substantially greater enrollment than the others and was the only one located in a purely urban setting. Another district was located on an urban-suburban boundary, whereas the remaining districts were located in suburban settings. The largest district also showed the greatest diversity in student ethnicity and the highest percentages of students eligible for a free or reduced-price lunch, nonnative English speakers, and special education students. In contrast, students in the remaining districts were predominately White native English speakers, most of whom were not eligible for a free or reduced-price lunch, and not involved in Special Education. The result of our purposive sampling is a dichotomy in enrollment and student characteristics in

Table 1
Location, Curriculum, Rationale, Assessment, and Enrollment for Sampled Districts

District	Geographic location	High school curriculum assessed in this study	Rationale for choice of district	Assessment used	Enrollment
A	Urban suburban boundary	MMOW	High fidelity of implementation with much parental support	SAT-9	11, 675
B	Urban	IMP	Large urban population—variable implementation	SAT-9	48,850
C	Suburban	CMIC	Wholesale adoption sabotaged by a few faculty dissenters and parents	SAT-9	3,466
D	Suburban	CMIC	Wholesale adoption with district authorized supplements	SAT-9	6,768
E	Suburban	CMIC	Enthusiastic adoption	SAT-9, NSREM	8,625
F	Suburban	MMOW	Partial adoption	NSREM	7,528

Note. SAT-9 = Open Ended and Multiple Choice subtests, NSREM = New Standards Reference Examination in Mathematics.

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

our samples of students and classrooms that we believe is largely representative of that existing in the target populations.

Data Collection

Cross-sectional data for two cohorts of 11th-grade students were gathered, one in Spring 2001 and the other in Spring 2002. There were no theoretical reasons to consider the cohorts of 11th-grade students tested as different. The results of descriptive analyses, as well as regression analyses including HLM in which cohort served as a predictor, indicated that there were no empirical reasons to treat the cohorts as separate. As a result, they were combined into a single group for analysis purposes. The initial samples for this study consisted of 972 high school students in 27 classrooms using a *Standards*-based curriculum; however, missing data reduced these sample sizes, especially for the HLM, which was based on 25 classrooms and 560 students. Table 1 shows that District F, a very affluent district, provided only criterion-referenced test data and hence was not included in the HLM analyses, which were based on standardized achievement test data.

A nonexperimental design was used in which students were considered to be clustered (nested) within classrooms, which in turn were clustered within school districts. Information was obtained for each level of clustering, but the focus was on students and classrooms. The lack of experimental manipulation means that study results support inferences about relationships among variables and their magnitude but do not generally support strong causal inferences (Rosenbaum, 2002, pp. 342–344).

This research project began with questions from districts, teachers, and parents concerning achievement of students enrolled in *Standards*-based classes in schools for which the $(MASP)^2$ LSC provided professional development. Our partner school districts wanted to assess students with instruments having national norms, despite the fact that most of these instruments were not designed to assess the effects of *Standards*-based curricula. After reviewing several instruments from national publishers and consulting with our partner districts, we narrowed the list to two: the Stanford Achievement Test, Ninth Edition (SAT-9) developed by Harcourt Brace Educational Measurement and the New Standards Reference Examination in Mathematics (NSREM) developed by the Learning Research and Development Center of the University of Pittsburgh and the National Center on Education and the Economy. Both sets of tests were supplied and scored by Harcourt Brace, and were administered under the supervision of project staff. The NSREM was designed to reflect *Standards*-based curricula, whereas the SAT-9 is a traditional standardized norm-referenced test that was not specifically designed to be sensitive to *Standards*-based curricula.

Specifically, the NSREM is a criterion-referenced test that assesses performance in three areas: Mathematical Skills, Mathematical Concepts, and Problem Solving. The performance levels are derived from national standards developed by a conglomerate of assessment-related organizations (Wiley & Resnick, 1998). The content and process areas assessed include number and operations, geometry and

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

measurement, algebra and function, mathematics skills, problem solving and reasoning, and mathematical communication.

Operationally, the first section of the NSREM in mathematics is composed of 20 multiple choice questions, which are a subset of SAT-9 multiple-choice test items, enabling this portion of the test to be compared to national norms. In addition, the first section contains short tasks. The student has 20 minutes to complete the multiple-choice questions and 35 minutes for the short tasks. Students then spend 55 minutes on the second section, which comprises long and medium length tasks. The third section requires an additional 55 minutes to complete and covers both short and long tasks. Short tasks are constructed-response items, whereas the medium and long tasks require detailed answers and are considered extended-response items.

The NSREM test reports student scores in a number of different ways, but we chose to examine student performance levels categorized as *achieved with honors*, *achieved the standard*, *nearly achieved*, *below standard,* and *little achievement*. Sample items for the NSREM are located in Appendix A.

The mathematics portion of the SAT-9 has two subtests. The Open Ended subtest consists of nine items with a total raw score value of 27 points. These items require students to construct extended responses and explain their solutions. "Although each item in the mathematics subtests can be solved by reasoning and an understanding of general mathematics, topics from algebra, geometry, trigonometry, statistics, discrete mathematics, and pre-calculus are included" (Harcourt Brace, 1996, p. 27). The Multiple Choice subtest consists of 30 items, each contributing 1 raw score point, which require students to solve problems set within real world and symbolic mathematical contexts. Calculators are allowed on both subtests, which each take two 50-minute time periods to administer. Sample items for the SAT-9 subtests are in Appendix B.

The SAT-9 reports both scaled scores and normal curve equivalents (NCEs). By definition, NCEs are normalized standard scores with a mean of 50 and a standard deviation of 21.06. The standard deviation of 21.06 was chosen so that NCEs of 1 and 99 are equivalent to percentiles of 1 and 99. (Stanford Technical Data Report, 1997). Although NCEs and percentiles are similar they are not identical. The data analysis results are generally reported in NCEs because of their familiarity and interpretability.

Table 2 presents information aligning NCE scores and SAT-9 percentiles with their corresponding scaled and raw scores. In the middle of the distribution of raw scores, a change of a single point (e.g., from 13 to 14 on Open Ended) results in a change of 12 percentile points and a change of about 6 NCE and scaled score points. Similarly, in the tails of the distribution of raw scores, a change of 1 point corresponds to a change of 3 to 5 percentile points and as many as 8 scaled score points. Thus, small changes in raw scores on the SAT-9 produce correspondingly larger changes in percentiles and scaled scores.

Teacher Professional Development

Teachers participated in three types of professional development. In one type, teachers participated in 2 weeks (approximately 80 hours) of summer training

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

Table 2
Selected SAT-9 Percentiles, NCEs, Scaled Scores, and Raw Scores (Spring 11th Grade Norms)

Open Ended				Multiple Choice			
%tile	NCE	SS	Raw	%tile	NCE	SS	Raw
94	82.7	714	20	98	93.3	781	26
90	77.0	704-705	19	90	77.0	750-751	22
85	71.9	696-697	18	85	71.8	740-741	21
60	55.3	672	14	61	55.9	710	15
48	48.9	666	13	56	53.2	705	14
39	44.1	661	12	51	50.5	700	13
13	26.3	636	7	17	29.9	671	8
10	23.0	631	6	12	25.3	665	7
7	18.9	625	5	8	20.4	657	6

Note. %tile = percentile, NCE = normal curve equivalent, SS = scaled scores.

related to a particular *Standards*-based curriculum. This usually entailed working through activities of the curriculum while teaching strategies were modeled by experienced $(MASP)^2$ staff members. In addition, mathematical content discussions were held each day, which were integrated into the work with student activities. In the second type, teachers participated in a series of sessions (approximately 30 hours) during the school year that focused on more general topics such as facilitating cooperative learning in mathematics classrooms, current research on the brain and its implications for mathematics classroom instruction, meetings with teachers and administrators to discuss administrative issues, and meetings with counselors relating to the scheduling of students. In the third type, $(MASP)^2$ employed district personnel experienced in the curriculum to serve during the school year as mentors to teachers newly implementing *Standards*-based curricula. The 20-hour mentoring component consisted primarily of classroom observations followed up with one-on-one debriefings and, in some cases, demonstration lessons. High school teachers in this study had completed an average of 144 professional development hours over a 3-year period.

An important consideration in a study examining the effectiveness of *Standards*-based curricula is the fidelity with which these curricula were implemented. Briars and Resnick (2000) determined that schools who implemented the Everyday Mathematics curriculum, as intended by the authors, scored higher on skills, problem solving, and concepts on the New Standards Reference Examination. *Standards*-based teaching was related positively to student achievement, especially when accompanied by a *Standards*-based curriculum (McCaffrey et al 2001). Weiss, Banilower, Overstreet, and Soar (2002) found that inquiry-oriented teaching practices were more prevalent when *Standards*-based classrooms were compared to traditional classrooms.

In a retrospective study such as the one reported here, assessing the fidelity of implementation can be difficult for a variety of reasons but is important because

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

of variation among teachers. Reys & Reys (2004, personal communication) found that teaching strategies consistent with the *Standards*-based text authors' suggestions were used from 10% to 90% of the time in the *Standards*-based classrooms they observed.

One factor promoting fidelity of implementation is that teachers whose classrooms were used in the present study participated in the $(MASP)^2$ professional development activities linked to *Standards*-based instruction. Other evidence of fidelity was provided by $(MASP)^2$ mentors assigned to provide an informal assessment of their teacher's implementation level in all classrooms between 1997 and 1999 and by information about individual teachers' classrooms as possessed by district evaluation personnel and district curriculum directors. Although no set of efforts can guarantee full fidelity, these considerations have helped to document that *Standards*-based curricula were generally implemented faithfully in our $(MASP)^2$ project classrooms.

Data Analyses

We used two methods to study students' achievement patterns. First, descriptive analyses were used to examine district- and classroom-level characteristics and achievement. As noted above, districts were concerned about how their students' scores compared to nationally normed groups, and we compared each district's mathematics performance to norms established by the test publisher. Second, we used HLM to model variation in mathematics achievement using both student- and classroom-level predictors. Student-level predictors included SES (not eligible for a free or reduced-price lunch, eligible), gender, and attendance, whereas classroom-level predictors included type of *Standards*-based curriculum (CMIC, IMP, MMOW), ethnic composition, and SES (percentage of students eligible for a free/reduced price lunch). Students were treated as clustered within classrooms (too few districts were tested to include this level of clustering). In a few cases, a single teacher taught two classes of students. To avoid dependency among classrooms with the same teacher we collapsed these two groups and henceforth considered them to be one classroom.

As other studies have noted (Begle, 1973; Reys, Reys, Lapan, Holliday, & Wasman, 2003; Riordan & Noyce, 2001), prior achievement in mathematics is an important predictor of student achievement. Accordingly, we needed a measure of prior achievement for use in several statistical analyses. All districts are mandated by the state to administer the mathematics portion of the Minnesota Basic Skills Test (MBST) in 8th grade, and the mathematics scores of this test, which were available in an NCE metric, served as our measure of Prior Mathematics Knowledge for both the student- and classroom-level regression models.

Background student data (e.g., ethnicity, attendance, student eligibility for a free or reduced-price lunch, English language learner status) were gathered from each district. As noted above, because of missing data most of the statistical analyses were based on fewer classrooms and students (25 and 560, respectively) than those orig-

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

inally sampled (27 and 972, respectively), with a substantial percentage of the missing values occurring for Prior Mathematics Knowledge and the SAT-9 subtests. The higher percentage of missing data for these variables is at least in part due to the 3-year gap between when these measures were administered (8th grade, 11th grade). Among the most plausible explanations for the greater percentage of missing scores for the Prior Mathematics Knowledge measure are that some students transferred into a district after completing 8th grade elsewhere and subsequently provided SAT-9 scores but no Prior Mathematics Knowledge scores. Similarly, some students who provided a prior mathematics score transferred out of a district before SAT-9 data were collected in 11th grade (information on student transfers was not available). It is also likely that some students simply failed to sit for these tests.

Using available data, we found no evidence that students with missing values differed systematically from those providing data (e.g., that a disproportionate number of missing values were linked to nonnative English speakers or to members of a particular ethnic group). As such discrepancies did not occur, we concluded that there was no clear evidence of bias. However, identifying bias is generally quite difficult and our findings should be interpreted in light of this possibility.

RESULTS

Method 1: Descriptive Summaries of District and Classroom Performance

SAT-9 descriptive statistics for the five districts providing SAT-9 data are presented in Table 3. Perhaps the most striking feature of Table 3 is that 9 out of 10 SAT-9 means were above 50, the national norm for NCE units for the Open Ended and Multiple Choice subtests. This indicates that the performance of *Standards*-based students was generally above average as determined by the test publisher's cutoff. It is also apparent that District E (suburban) had, on average, the highest NCE scores on both the Open Ended and Multiple Choice subtests and District B (urban) the lowest. Four of the five district means for Prior Mathematics Knowledge exceeded 50, with District B falling well below this benchmark. Overall, 16 NCE points separated the highest and lowest performing school districts

Table 3
Descriptive Statistics for the SAT-9 Subtests by District

	Prior Mathematics Knowledge			Open Ended			Multiple Choice		
District	*n*	*M*	*SD*	*n*	*M*	*SD*	*n*	*M*	*SD*
A	284	55.2	14.1	380	63.0	20.4	372	54.3	15.1
B	142	41.7	15.3	162	52.1	23.0	171	45.7	17.0
C	44	52.5	14.4	50	62.4	19.0	50	59.9	14.2
D	220	64.0	17.9	163	59.0	24.1	168	57.7	17.2
E	27	57.6	15.7	21	67.7	21.2	21	65.1	17.6
Total	717	55.2	17.5	776	60.0	22.1	782	53.8	16.7

Note. Means (*M*) and standard deviations (*SD*) are reported in NCE units, *n* = sample size.

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

on Open Ended, and 19 points on Multiple Choice. There was also considerable variability among districts in average Prior Mathematics Knowledge (22 points).

The districts also differed in student characteristics, with District B showing the highest percentage of nonwhite students (just under 60%), which included African American (35.3%) and Asian (20.6%) students who produced similar means on the Open Ended (43.9 and 46.7, respectively) and Multiple Choice subtests (41.7 and 42.1, respectively). The small number of Hispanic students (11) across the five districts led us to exclude them from all analyses. District B also had the highest percentage of students eligible for a free or reduced-price lunch (low SES, just under 46%) and nonnative English speaking students (just over 12%), with the remaining districts showing substantially smaller percentages for these variables ranging between 0 and 10%. The percentage of students in special education in our sample across the districts ranged from 0% to 5.5%.

Table 4 reports average achievement for various student subgroups by district. It is apparent that low SES, African American, nonnative English speakers, and special education students performed below their counterparts, although there were a few exceptions in one suburban district.

The Open Ended, Multiple Choice, and Prior Mathematics Knowledge tests also showed similar patterns across various student subgroups as seen in Table 5. The difference in Prior Mathematics Knowledge means in Districts A and B favored the high SES group over students in the low SES group (9 NCE points and 8 points, respectively), 11 and 13 points for the Open Ended subtest, and 10 and 9 points for Multiple Choice. In most cases, the mean of the low SES group was below 50.

White students in District B scored on average 6–8 points higher than African American and Asian students on Prior Mathematics Knowledge, 8–15 points higher on the Open Ended subtest, and 7–8 points higher on Multiple Choice. White students scored on average 3 and 7 points lower than African American students on Prior Mathematics Knowledge in Districts C and D, respectively. However, White students scored 5–6 points higher than African American students in these districts on the Open Ended and Multiple Choice subtests.

In District B native English speakers scored approximately 9 NCE points higher than nonnative speakers on Prior Mathematics Knowledge, 22 points higher on Open Ended, and 5 points higher for Multiple Choice. The means for nonnative speakers were well below 50.

Additional descriptive statistics for high school students including effect sizes (*d*) appear in Table 5, which were computed across districts. For example, low SES students scored on average .68 standard deviations lower on the Open Ended subtest than high SES students. One obvious finding in Table 5 is the lower average performance for low SES, nonwhite, urban, nonnative English speakers, and special education students. These effect sizes ranged from .11 to 2.00 standard deviation units lower than the comparison group. An examination of test performance by classroom (see Table 6) provided evidence of substantial variation in achievement.

Finally, we examined average performance for the three *Standards*-based curricula in Table 7. On the Open Ended subtest, students in MMOW produced a

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

Table 4
Average Achievement for Various Subgroups by District

District		SES		Ethnicity			Language	
		High	Low	White	African American	Asian American	English speaker	Nonnative speaker
A	Prior Knowledge	56.0(258)	47.0(26)	55.2(260)	60.8(17)	—	55.1(282)	—
	Open Ended	63.8(351)	52.7(29)	63.6(351)	57.2(20)	—	63.1(378)	—
	Multiple Choice	55.1(343)	44.8(29)	54.8(344)	49.8(19)	—	54.3(370)	—
B	Prior Knowledge	46.1(67)	37.8(75)	45.7(54)	38.4(54)	40.7(34)	43.1(120)	34.3(22)
	Open Ended	57.0(101)	43.9(61)	59.3(80)	43.9(49)	46.8(33)	54.7(143)	32.8(19)
	Multiple Choice	49.2(103)	40.4(68)	49.7(84)	41.7(53)	42.1(34)	46.3(152)	40.7(19)
C	Prior Knowledge	53.8(42)	—	51.7(43)	—	—	52.5(44)	—
	Open Ended	63.1(48)	—	61.7(49)	—	—	62.4(50)	—
	Multiple Choice	60.9(47)	—	59.3(49)	—	—	59.9(50)	—
D	Prior Knowledge	64.3(210)	56.6(10)	64.0(201)	66.7(17)	—	—	—
	Open Ended	59.2(157)	—	59.9(144)	52.5(19)	—	59.3(162)	—
	Multiple Choice	58.0(163)	—	57.3(150)	60.6(18)	—	57.8(167)	—
E	Prior Knowledge	58.6(26)	—	58.1(25)	—	—	57.6(27)	—
	Open Ended	68.3(20)	—	68.9(20)	—	—	67.7(21)	—
	Multiple Choice	65.6(20)	—	65.6(20)	—	—	65.1(21)	—

Note. Sample size is in ( ). — means that $n < 10$ cases were available, which was deemed too small for credible estimation of the mean.

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

Table 5
SAT-9 Subtest Achievement for Various Subgroups (Across Districts)

		Prior Knowledge				Open Ended				Multiple Choice			
		n	*M*	*SD*	Effect size	*n*	*M*	*SD*	Effect size	*n*	*M*	*SD*	Effect size
SES	High	603	57.8	16.6		677	61.8	21.5		676	55.6	16.0	
	Low	114	41.3	15.0	1.01*	99	47.1	21.8	0.68*	106	42.2	16.4	0.84*
Ethnicity	White	583	57.2	16.3		644	62.2	21.2		647	55.4	16.1	
	African American	90	48.7	20.9	0.42*	90	49.2	25.3	0.52*	92	47.8	18.3	0.46*
	Asian American	44	41.3	15.0	0.98*	42	48.0	18.4	0.68*	43	42.8	15.0	0.78*
Gender	Male	340	56.7	17.0		364	61.3	22.5		370	55.3	16.2	
	Female	377	53.8	17.8	0.16*	412	58.8	21.7	0.11	412	52.5	17.0	0.17*
Location	Suburban	575	58.5	16.3		614	62.0	21.4		611	56.1	15.9	
	Urban	142	41.7	15.3	1.04*	162	52.1	23.0	0.44*	171	45.7	17.0	0.64*
Language status	English speaker	693	55.8	17.1		754	60.8	21.8		760	54.2	16.6	
	Nonnative speaker	24	36.8	16.8	1.11*	22	32.5	13.8	2.00*	22	41.6	14.6	0.76*
Special education	Nonspecial ED	687	55.7	17.4		749	60.5	21.9		757	54.2	16.6	
	Special ED	30	42.7	13.2	0.75*	27	45.9	24.2	0.66*	25	42.4	15.5	0.71*

Note. $*p < 0.05$ (*t* tests). *n* = sample size, *M* = mean, *SD* = standard deviation.

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

Table 6
Prior Mathematics Knowledge and Classroom Outcomes Expressed in Normal Curve Equivalents (NCEs)

Class	Prior Mathematics Knowledge	Open Ended	Multiple Choice	Location	Curriculum
1	76.3	80.3	70.8	Suburban	CMIC
2	56	68.2	56.9	USB*	CMIC
3	80.8	67.9	67.8	Suburban	CMIC
4	57.6	67.7	65.1	Suburban	CMIC
5	64	65.7	70.3	Suburban	CMIC
6	51.5	60.7	52.9	Suburban	CMIC
7	47.2	60.5	55.1	Suburban	CMIC
8	84.8	55.9	53.9	Suburban	CMIC
9	57.1	54.7	58.4	Suburban	CMIC
10	55.2	52.5	53.4	Suburban	CMIC
11	39.3	61.6	47.6	Urban	IMP
12	40.6	59.7	50.9	Urban	IMP
13	51.9	57.7	50.4	Urban	IMP
14	44.6	55.9	37.5	Urban	IMP
15	51.8	54	54.5	Urban	IMP
16	50.2	46	41.8	Urban	IMP
17	48.4	45.5	42.7	Urban	IMP
18	37.8	44.8	39.2	Urban	IMP
19	35.8	37.6	42.9	Urban	IMP
20	39.8	31.8	49.1	Urban	IMP
21	55.1	68.1	51.6	USB*	MMOW
22	56.8	65.9	54.2	USB*	MMOW
23	53	60.5	53.6	USB*	MMOW
24	54.3	60.2	53.3	USB*	MMOW
25	53.4	54.6	55.8	USB*	MMOW

Note. *USB is Urban-Suburban Boundary.

Table 7
Prior Mathematics Knowledge, Open Ended, and Multiple Choice Achievement for the Standards-based Curricula (Across Districts)

	Prior Mathematics Knowledge			Open Ended			Multiple Choice		
Curriculum	*n*	*M*	*SD*	*n*	*M*	*SD*	*n*	*M*	*SD*
IMP	142	41.7	15.3	162	52.1	23.0	171	45.7	17.0
CMIC	291	61.6	17.6	234	60.5	22.9	239	58.7	16.7
MMOW	284	55.2	14.1	380	63.0	20.3	372	54.3	15.4

Note. *n* = sample size, *M* = mean, *SD* = standard deviation.

mean of 63 NCE units, well above the publisher's proficiency cutoff of 50 and higher than those in CMIC (60.5) or IMP (52.1). For the Multiple Choice subtest, CMIC students scored highest on average (58.7), followed by MMOW (54.3) and IMP (45.7). This pattern also emerged for Prior Mathematics Knowledge, with CMIC students scoring on average the highest (61.6), followed by MMOW (55.2) and IMP

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

(41.7). One-way, fixed-effects ANOVA revealed that the three curriculum groups differed significantly for each of the outcome variables appearing in Table 6. Using Fisher's eta-squared ($\hat{\eta}^2$) as an effect size, .17 or 17% of the variation in Prior Mathematics Knowledge was a function of curriculum, whereas for Open Ended and Multiple Choice the $\hat{\eta}^2$ values were .04 and .08, respectively. Thus, the largest differences between curriculum groups appeared for Prior Mathematics Knowledge.

The fact that the Prior Mathematics Knowledge test was administered in the 8th grade and the Open Ended and Multiple Choice subtests in the 11th grade, and all measures are reported in NCE units, allows a rough comparison of change over time by curriculum. Of course, precise comparisons over time require the use of scale scores and the same students measured repeatedly, and we offer the following only as a potentially interesting facet of the data.

As seen in Table 7, IMP students scored on average the lowest on the Prior Mathematics Knowledge test (41.7) and above the proficiency cutoff of 50 on the Open Ended test administered in the 11th grade (52.1). If these students had shown no change in their mathematics achievement, we might expect the two NCE means to be similar, which was not the case. CMIC students scored about the same on average on these two tests (61.6 and 60.5, respectively), whereas *MMOW* students showed positive change (Prior Mathematics Knowledge = 55.2, Open Ended = 63). For Multiple Choice, IMP students showed on average modest growth between 8th grade and 11th grade, CMIC showed a modest decline, and MMOW students stayed about the same.

The findings above need to be considered in light of the fact that all IMP classrooms were in the urban district, which generally produced lower average test scores than the suburban classrooms. Thus, comparisons of student performances linked to IMP and the other *Standards*-based curricula are confounded with classroom location. We discuss this confounding further when presenting our HLM results.

In sum, there is ample evidence of variability among the school districts and classrooms in achievement as well as background variables. The suburban districts, including the district on the urban-suburban boundary, tended to have far smaller percentages of nonwhite, low SES, and nonnative English speakers. The distribution of special education students showed little relationship with the suburban or urban location of districts. There was also evidence that students in the CMIC and MMOW curricula scored on average well above those in IMP in mathematics achievement.

New Standards Reference Exam in Mathematics

The NSREM was only administered in suburban Districts E and F, and we therefore have NSREM data on only a relatively small number of students: four classrooms ($n = 92$) in District E and one classroom in District F ($n = 22$). These findings are reported as percentages of students who achieved in each of five categories, as denoted in Table 8.

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

Table 8
Percentage and Number of Students Meeting New Standards Achievement Level—District E and F

	Mathematical skills		Mathematical concepts		Problem solving	
Achievement level	District E &F	National	District E &F	National	District E &F	National
Achieved with honors	79% (90)	11%	49% (56)	4%	18% (20)	2%
Achieved the standard	19% (22)	39%	31% (35)	10%	40% (46)	7%
Nearly achieved	1% (1)	16%	11% (13)	16%	18% (21)	6%
Below the standard	1% (1)	25%	8% (9)	37%	20% (23)	41%
Little achievement	0% (0)	9%	1% (1)	33%	4% (4)	44%

Note. Total sample size $n = 114$.

The percentage of students who achieved the highest category (*Achieved with Honors*) in Mathematical Skills was 79%, which contrasts sharply with the 11% in this category based on the national norm group. Ninety-eight percent of students achieved the standard (top two categories) compared to 50% in the norming group. The pattern was similar for Mathematical Concepts and Problem Solving.

Method 2: HLM Analyses of Student and Classroom Data

The Open Ended and Multiple Choice outcomes were analyzed separately with HLM following the methods described in Raudenbush and Bryk (2002) and using the HLM6 software (Raudenbush, Bryk, & Congdon, 2005). Treating students as clustered within classrooms permitted within-classroom dependency among student mathematics test scores to be modeled, and allowed both student- and classroom-level data to be analyzed simultaneously. HLM also allowed us to assess whether the impact of the three *Standards*-based curricula on achievement was similar with various background variables taken into account.

Student-level regression models containing Prior Mathematics Knowledge, attendance, SES, and gender were fitted to the data for each classroom. These predictors have been shown to frequently impact mathematics achievement (International Association for the Evaluation of Educational Achievement, 2003; Post, Cutler, Davis, Anderson, Maeda, Kahan et al., in press; Reys et al., 2003; Sirin, 2005; Tate, 1997), and largely served as control variables to enhance inferences. For each outcome (Open Ended, Multiple Choice), three models were fitted.

First an unconditional model of the form

$$(1) \qquad \begin{aligned} Y_{ij} &= \beta_{0j} + r_{ij} \\ \beta_{0j} &= \gamma_{00} + u_{0j}, \end{aligned}$$

was fitted, where Y_{ij} is the mathematics score of the ith student in the jth classroom, β_{0j} is the average mathematics score (intercept) for the jth classroom, γ_{00} is the average mathematics performance across classrooms, r_{ij} is a student-level residual,

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

and u_{0j} represents the unique effect of the jth classroom. The unconditional model results tell us whether average outcomes differ across classrooms.

Next we fitted a student-level model of the form

$$Y_{ij} = \beta_{0j} + \beta_{1j}(SES_{ij} - \overline{X}_1) + \beta_{2j}(Attendance_{ij} - \overline{X}_2) + \beta_{3j}(Gender_{ij} - \overline{X}_3) + \beta_{4j}(Prior_{ij} - \overline{X}_4) + r_{ij}, \quad (2)$$

where β_{1j} is the student-level slope capturing the effect of SES (0 = not eligible for a free or reduced price lunch, 1 = eligible) on mathematics performance with other predictors held constant in the jth classroom, $\overline{X}_1$ is the grand mean for SES, and SES_{ij} is the SES value of the ith student in the jth classroom. We also tested whether slopes for the predictors varied across classrooms. Only attendance slopes varied significantly across classrooms, and we constrained the variances for the SES, gender, and Prior Mathematics Knowledge slopes to zero in the modeling. Grand-mean-centering of student-level predictors ensured that intercepts were adjusted for student-level predictors.

Classroom-level predictive models for intercepts and attendance slopes were developed. That is, we attempted to model variation in intercepts and attendance slopes across classrooms using classroom-level predictors. We again relied on previous findings to guide the selection of key predictors, which included class SES (percentage of students eligible for a free or reduced-price lunch in a classroom) and average Prior Mathematics Knowledge in a classroom. Other classroom predictors that we examined were the effect of different concentrations of various ethnic groups, nonnative English speakers, special education students, female students, and professional development hours of the teacher. The professional development hours linked to a teacher and the percentage of females in the classroom-level model were not statistically significant in any of the analyses, and refitting the model after omitting these variables did not change the findings in an important way. The final classroom-level model for intercepts was

$$\begin{aligned}\beta_{0j} = {} & \gamma_{00} + \gamma_{01}(Class\ SES_j - \overline{W}_1) \\ & + \gamma_{02}(Class\ Nonnative\ Speaker_j - \overline{W}_2) \\ & + \gamma_{03}(Class\ Special\ Education_j - \overline{W}_3) \\ & + \gamma_{04}(Class\ African\ American_j - \overline{W}_4) \\ & + \gamma_{05}(Class\ Prior_j - \overline{W}_5) + \gamma_{06}(IMP - \overline{W}_6) \\ & + \gamma_{07}(CMIC - \overline{W}_7) + u_{0j},\end{aligned} \quad (3)$$

where γ_{01} is the classroom level slope capturing the effect of class SES (percentage of low SES students) on average mathematics performance, $\overline{W}_1$ is class SES averaged across classrooms, and class African American is the percentage of African American students in a classroom. Model (3) was also used to predict variation in attendance slopes.

The predictors IMP (1 = IMP curriculum used in classroom, 0 = not used) and CMIC (1 = CMIC curriculum used in classroom, 0 = not used) captured which of

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

the three *Standards*-based curricula a student experienced. However, all of the IMP classrooms were in the urban district, whereas CMIC and MMOW classrooms were distributed over the remaining suburban districts. Thus, comparisons of IMP and non-IMP classrooms also potentially reflect differences between urban and suburban classrooms. The confounding of classroom location and which *Standards*-based curriculum was used in a classroom is not easily disentangled, but the fact that HLM statistically controlled for various factors distinguishing urban and suburban classrooms (percentages of low SES students, nonnative English speakers, African American students, etc.) allows us to argue that achievement comparisons between IMP and non-IMP classrooms likely reflect different curricula rather than being solely attributable to classroom location.

The deviance test described in Raudenbush and Bryk (2002, pp. 60–61) was used to test for model fit, allowing us to discriminate among models with more or less explanatory power. Model fitting was followed by extensive model checking to help to ensure validity of inferences. Cases in which normality, homoscedasticity, or linearity appeared to be suspect were examined in detail, and various remedies (e.g., modeling unequal classroom variances) employed. The analyses reported below are based on fitted models in which these assumptions appeared to be at least approximately satisfied. Two classrooms with four or less students were omitted because the student-level model in (2) could not be fitted. The data for these classrooms, although quite sparse, could have been used to estimate fixed effects and variance components using the HLM6 software but could not be used in testing variance components. This led us to omit these two classrooms from the HLM analyses. Missing data resulted in the loss of other cases, leaving 25 classrooms from 5 districts and 560 students for these analyses.

An initial difficulty with several of the classroom-level predictor variables, such as the percentage of nonnative English speakers in a classroom, was their ragged and discontinuous nature. For example, 52% of the classrooms had no nonnative English speakers, 28% of the classrooms had values between 1%–8%, and, at the other end of the distribution, five of the classrooms (20%) had values well above 8%. We explored various transformations of these variables with the goal of representing their variation in a more succinct form, and subsequently polytomized the distributions into quartiles. Thus, each of the classroom predictors in (3), with the exception of Prior Mathematics Achievement, which was not transformed, and percentage nonnative English speakers, which was trichotomous because of its distribution across classrooms, were transformed to a scale in which each was represented with one of four values corresponding to the quartiles. These values are presented in Table 9.

The HLM cross-sectional results are summarized in Table 10. Several findings emerged across the Open Ended and Multiple Choice subtests. First, there was significant, but not overwhelming, between-classroom variation. Approximately 16% of the variation among Open Ended means was accounted for by the classroom that students were in; for Multiple Choice, 22% of the variation in classroom means was accounted for by classroom. Second, at the student level, Prior

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

Table 9
Definitions of Classroom Level Predictors

Variable	Quartile	*n*	Range (*R*)
Class SES (low)	1	6	$R \leq 5.12\%$
	2	6	$5.12\% < R \leq 8.59\%$
	3	7	$8.59\% < R \leq 40.38\%$
	4	6	$R > 40.38\%$
Class nonnative speaker status	1	13	$R = 0\ \%$
	2	6	$0\% < R \leq 8.01\%$
	3	6	$R > 8.01\%$
Class special education	1	11	$R = 0\ \%$
	2	1	$0\% < R \leq 2.08\%$
	3	7	$2.08\% < R \leq 5.56\%$
	4	6	$R > 5.56\%$
Class African American	1	6	$R \leq 4.21\%$
	2	6	$4.21\% < R \leq 7.81\%$
	3	7	$7.81\% < R \leq 25.14\%$
	4	6	$R > 25.14\%$
Class Asian American	1	9	$R = 0\%$
	2	3	$0\% < R \leq 3.7\%$
	3	7	$3.7\% < R \leq 11.66\%$
	4	6	$R > 11.66\%$
Professional development hours	1	6	$R = <$ 104 hours
	2	7	$104 < R \leq 156$
	3	7	$156 < R \leq 188$
	4	5	$R > 188$
Class Prior Mathematics Knowledge	—	25	Classroom average Prior Mathematics Knowledge

Note. Class Prior Mathematics Knowledge was not polytomized. *n* = number of classrooms.

Mathematics Knowledge was a statistically significant predictor in every model, although its effect expressed in NCE units was quite modest (< 1). Student-level SES was statistically significant in the model for Multiple Choice and demonstrated a moderate effect on mathematics performance (–4.74), meaning that, with the other predictors held constant, low SES students tended to score almost 5 NCE points less than high SES students. Gender and attendance were not statistically significant student-level predictors. However, there was a significant cross-level interaction involving the attendance slopes and the percentage of African American students (expressed in quartiles) in a classroom (1.21), indicating that the effect of attendance on Multiple Choice scores increased as the concentration of African American students in a classroom increased.

Third, the Multiple Choice results showed fewer significant effects than those for the Open Ended subtest. Fourth, statistically significant contextual effects for Prior Mathematics Knowledge on the Open Ended and Multiple Choice subtests suggested that the effect of Prior Mathematics Knowledge on mathematics achievement was stronger at the classroom level than at the student level. Fifth, no differ-

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

Table 10
HLM Results

Dependent variable	HLM results
Open Ended	1. There was significant between-classroom variation in means (16%). 2. The only significant student-level effect was for Prior Mathematics Knowledge (.68). Open Ended intercepts and attendance slopes varied significantly across classrooms. 3. There was a small but significant contextual effect for Prior Mathematics Knowledge (–.57), meaning that the effect of this factor on Open Ended scores differs at the student and classroom levels. The evidence suggests that the effect is stronger at the classroom level. 4. None of the predictors in equation (3) were significant predictors of classroom intercepts. Although students in the IMP curriculum scored on average 7.7 points lower than those in non-IMP curricula with other predictors held constant, this effect was not significant ($p = .19$). In a similar way, students in the CMIC curriculum scored on average 5.6 points lower than those in non-CMIC curricula; this effect was not significant ($p = .10$). 5. The predictive models for attendance slopes had one significant effect. Increasing concentrations of African American students in a classroom were associated with a stronger effect of attendance on Open Ended scores (1.21).
Multiple Choice	1. There was significant between-classroom variation in means (22%). 2. The largest (significant) student-level effect was for SES (–4.72), meaning that, with other predictors held constant, low SES students tended to score just under 5 NCE points lower than high SES students. Prior Mathematics Knowledge was also a significant student-level predictor (.50). 3. There was a small but significant contextual effect for Prior Mathematics Knowledge (–.37), meaning that the effect of this factor on Open Ended scores differs at the student and classroom levels. The evidence suggests that the effect is stronger at the classroom level.

ence among the three *Standards*-based curricula emerged on the SAT-9 subtests when other variables were taken into account.

DISCUSSION

This was not (and could not be) a true experimental study (Campbell & Stanley, 1966), because the retrospective nature of the study means that students and teachers could not have been assigned at random to urban or suburban classrooms or to a particular *Standards*-based curriculum, implying that strong causal inferences are not warranted. We attempted to describe patterns of student achievement from a number of perspectives, including HLM analyses, which permitted us to view classroom and student data simultaneously and to model within-classroom dependencies.

As noted, the districts were selected for different reasons. One was characterized by enthusiastic district-wide adoption, whereas in another, there were active efforts to undermine the use of *Standards*-based curricula. Our remaining districts

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

reflected everything in between these extremes. Both affluent suburban districts and one primarily low SES heavily multicultural urban district were represented. Our goal was to provide a view of student mathematics achievement in classrooms that had completed three levels of involvement (three Carnegie units) with *Standards*-based curricula.

The reader will note that we have used the 50th percentile (NCE mean = 50, and scaled scores at the 50th percentile) on the SAT-9 as the benchmark to which students' scores were compared. Why the 50th and not the 65th or 75th percentile? Surely the national mean is not an aspiration for all districts; for example, our high achieving suburban districts would be quite displeased were their students to achieve at this level. On the other hand, the 50th percentile is a realistic goal for our urban district, 73% of whose students (60% in our sample) are minority and which contains a relatively large group of low SES students. In the final analysis, we selected the 50th percentile because we felt it was an appropriate goal for our lower achieving districts, whereas, of course, also permitting us to gauge the degree to which our higher achieving districts exceeded (or did not) the national mean.

No single assessment can accurately assess all that a student knows and understands. This is especially true when asking whether a traditionally oriented standardized test is capable of assessing the efficacy of a *Standards*-based curriculum, given the wide disparity (a validity issue) between content and emphasis. Although there are a number of issues surrounding these tests, attempts such as this to assess traditional topics, and compare results to national norms, is nonetheless an important exercise. Such activity was consistent with, and useful to, our superintendents' perceived need to obtain this kind of information, primarily for parental consumption. The SAT-9 results contained no surprises: Our urban and boundary suburban district achieved at about the national norm, whereas our more affluent suburban districts scored higher. Procedural (Multiple Choice) scores lagged behind Open Ended results in *Standards*-based classrooms.

By 1998, District E, our highest achieving district, had implemented *Standards*-based curricula for *all* its students, adopting Everyday Mathematics (Bell, 1988–1996) at the elementary level, Connected Mathematics (Lappan, Fey, Fitzgerald, Friel, & Phillips, 1991–1997) at the middle grades, and CMIC at the high school level. The criticism from opponents of *Standards*-based curricula that high achieving students would be "held back" was not supported by the results that follow. As part of a district-wide self-assessment, a study distinct from the data included here, District E reported that the number of its high school students taking the BC level of the AP Calculus exam jumped from 50 to 67 between 1999 and 2003. The passing rate with a score of 3 or better during this period increased from 64% to 87%. For AB calculus, the number of students increased from 10 to 16, and there was an increase in the percentage with a grade of 3 or more from 30% to 81%. Similarly, the number of students taking the AP Statistics exam increased from 31 to 71 between 1999 and 2003. The passing rate, with a score of 3 or higher, during this period also increased slightly from 74% to 76%. In 2003, all high school students in District E had been exposed only to the CMP and CMIC programs since grade 6.

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

Such information is at odds with the negative and cautionary claims of the "mathematically correct" and other groups claiming that *Standards*-based curricula are inappropriate for higher achieving students. In this case, District E teachers, administrators, and most parents are strong in their support for this program and are resolute in their determination to continue with its use. In general, district teachers and administrators (and parents) need to recognize that there is emerging evidence that *Standards*-based curricula do not impede the mathematical performance and development of higher achieving students, and in some cases the curricula appear to have contributed to better student performance. As such evidence continues to mount, school personnel can become increasingly confident in their adoption and implementation of *Standards*-based mathematics curriculum materials with all their students.

Given results such as those obtained from the NSREM, and despite the fact that these were two high achieving districts to begin with, it is difficult to understand the ongoing concern by some that *Standards*-based students are not learning basic mathematical skills and are lacking important mathematical understandings. This issue has been an important motivator for this study, and we are pleased that additional light is being shed on the issue. Coupled with other results from similar studies (e.g., Senk & Thompson, 2003), perhaps it is now time to focus research activity on what mathematics *Standards*-based students are learning that traditional students are not, and vice versa. It will be prudent in future comparative studies to probe the depth of student understanding, their flexibility in accessing various mathematical processes and skills while engaged in problem solving, their realization of the scope of the mathematical connections made, and the ease and accuracy of their applicability to new situations.

Because clear evidence supporting alternatives to *Standards*-based curricula have not been published, one can conclude that opponents would have us "stay the course." Hiebert (2003) suggested that traditional approaches over the past 30 years have not produced acceptable results either in student achievement levels or in the percentages of students who continue the study of mathematics beyond minimum high school requirements. It is unfortunate that the political debates about the efficacy of the reformed mathematics curricula have for the most part unduly influenced research efforts to assess student achievement on traditional topics as measured by traditionally oriented standardized tests. This study was no exception.

Assessing reformed students' achievement on traditional topics is not the most interesting or ultimately the most important question. It is however an important component of an exploratory research agenda. NSF did not fund the development of thirteen complete curricula at the elementary, middle grades, and secondary level in the 1990s to simply increase achievement on existing standardized tests. The NSF initiative occurred because of an increasingly disparate achievement pattern between the "haves" and the "have nots"; because the half-life of mathematics students is 1 year, beginning in 8th grade; and because the United States as the world's technological leader is currently in serious jeopardy (Friedman, 2005). Further, advances in technology have made preoccupation with algorithmic procedures unrealistic. Evolving conditions in the workplace have expanded the need for a deeper under-

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

standing of probability, statistics, and the problem-solving power of topics in discrete mathematics, traditionally topics not significantly addressed at the secondary level. The traditional separation and isolated treatment of algebra, geometry, and trigonometry have predictably inhibited students' ability to consider problems from multiple perspectives. For example, spreadsheets and statistical regression are two of the most widely used mathematical tools in the business community, but these tools are not represented in traditional approaches.

The preponderance of the research to date has indicated that there are essentially no differences between *Standards*-based and traditional students' achievement on traditional topics, with the probable exception of paper-and-pencil calculation, and speed and accuracy with algorithmic procedures. Given the variety and extent of computer (and calculator) algebra systems, such hand manipulation is a skill of far less importance in the 21st century. This is not to say that such skills are unimportant but there must be additional dimensions to program design for the 21st century for the majority of our student population.

The results obtained here do not suggest that *Standards*-based curricula are the magical solution to ongoing and persistent complaints about students' lack of mathematical preparedness in traditional topical domains. It must be noted that this has been a recurring concern (complaint) for most of the last century. It seems prudent now to refocus research efforts to other and more important areas of investigation that will expand the domain of research to include the learning of probability, statistics, topics in discrete mathematics, and students' ability to solve nonroutine problems and apply mathematics prudently in contextual settings. A second and promising area for investigation will be students' performance in postsecondary mathematics and related coursework. It is time now to devote research energy to the more important question of what mathematics *Standards*-based students are learning that traditional students are not, and vice versa, and what ancillary attributes accompany such learning.

REFERENCES

Abeille, A., & Hurley, N. (2001). Final evaluation report: Mathematics modeling our world. Accessed on October 18, 2003, from http://www.comap.com/highschool/projects/mmow/FinalReport.pdf.

Begle, E. G. (1973). Some lessons learned by SMSG. *Mathematics Teacher*, *66*, 207–214.

Bell, M. (1988–1996). *Everyday mathematics*. Chicago: Everyday Learning Corporation.

Berlinghoff, W. P., Sloyer, C., & Hayden, R. W. (2000). Math connections: A secondary core curriculum (Years 1–3). Armonk, NY: It's About Time.

Briars, D., & Resnick, L. (2000). *Standards, Assessments—and What Else? The essential elements of* Standards-*based school improvement*. Unpublished manuscript. Retrieved October 15, 2003, from http://www.cse.ucla.edu/CRESST/Reports/TECH528.pdf

Bruner, J. S. (1964). The course of cognitive growth. *American Psychologist*, *19*, 1–15.

Bruner, J. S., Olver, R. R., & Greenfield, P. M. (1966). *Studies in cognitive growth*. New York: Wiley.

Campbell, D. T., & Stanley, J. C. (1966). *Experimental and quasi-experimental designs for research*. Chicago: Rand, McNally & Company.

Cochran, W. G. (1983). *Planning and analysis of observational studies*. New York: John Wiley.

Cichon, D., & Ellis, J. G. (2003). The effects of Math *Connections* on student achievement, confidence, and perception. In S. L. Senk and D. R. Thompson (Eds.), *Standards-based school mathematics*

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

curricula: What are they? What do students learn? (pp. 345–374). Mahwah, NJ: Lawrence Erlbaum Associates.

Coxford, A. F., Fey, J. T., Hirsch, C. R., Schoen, H. L., Burrill, G., Hart, E. W. et al. (1998). Contemporary mathematics in context: A unified approach (Courses 1–4). New York: Glencoe/McGraw-Hill.

Dewey, J. (1916). *Democracy and education.* New York: The Macmillan Company.

Dienes, Z. P. (1969). *Building up mathematics*. Hutchinson Educational, WI: London.

Feldt, L. S., Forsyth, R. A., Ansley, T. N., & Alnot, S. D. (1993). *Iowa tests of educational development (Forms K & L)*. Chicago: Riverside.

Fendel, D., Resek, D., Alper, L., & Fraser, S. (1997). Interactive mathematics program (Years 1–4). Berkeley, CA: Key Curriculum Press.

Friedman, T. L. (2005). *The world is flat: A brief history of the twenty-first century*. New York: Farrar, Straus, and Giroux.

Furth, H. G. (1970). *Piaget for teachers*. Upper Saddle River, NJ: Prentice-Hall.

Garfunkel, S., Godbold, L., & Pollak, H. (1998). Mathematics: Modeling our world (Courses 1–4). Cincinnati, OH: South-Western Educational Publishing.

Ginsburg, H., & Opper, S. (1988). *Piaget's theory of intellectual development* (2nd Ed.). Upper Saddle River, NJ: Prentice-Hall.

Harcourt Brace. (1996). *Stanford achievement test, ninth edition.* San Antonio, TX: Author.

Hedges, L.V., & Olkin, I. (1985). *Statistical methods for meta-analysis*. San Diego, CA: Academic Press.

Hiebert, J. (2003). What research says about the NCTM Standards. In J. Kilpatrick, W. G. Martin, & D. Schifter (Eds.), *A research companion to principles and standards for school mathematics* (pp. 5–24). Reston, VA: National Council of Teachers of Mathematics.

Huntley, M. A., Rasmussen, C. L., Villarubi, R. S., Sangtong, J., & Fey, J. T. (2000). Effects of *Standards*-based mathematics education: A study of the Core-Plus Mathematics Project algebra and functions strand. *Journal for Research in Mathematics Education, 31*, 328–361.

Kilpatrick, J. (2003). What Works? In S. L. Senk and D. R. Thompson (Eds.), *Standards-based school mathematics curricula: What are they? What do students learn?* (pp. 471–487). Mahwah, NJ: Lawrence Erlbaum Associates.

Kozulin, A. (1986). The concept of activity in Soviet psychology: Vygotsky, his disciplines and critics. *American Psychologist*, *41*, 264–274.

Lambdin, D. V., & Preston, R. V. (1995). Caricatures in innovation: Teacher adaptation to an investigation-oriented middle school mathematics curriculum. *Journal of Teacher Education*, *46*, 130–140.

Lappan, G., Fey, J. T., Fitzgerald, W. M., Friel, S. N., & Phillips, E. D. (1991-1997). *Connected mathematics*. White Plains, NY: Dale Seymour Publications.

Lesh, R. (1979). Mathematical learning disabilities: Considerations for identification, diagnosis, and remediation. In R. Lesh, D. Mierkiewicz, & M. Kantowski (Eds.), *Applied mathematical problem solving* (pp. 111–180). Columbus, OH: ERIC/SMEAC.

Lott, J. W., & Burke, M. (1996). Integrated mathematics: A modeling approach using technology (Levels 1–6). Boston: Pearson Custom Publishing.

Lott, J. W., Hirstein, J., Allinger, G., Walen, S., Burke, M., Lundin, M., et al. (2003). Curriculum and assessment in *SIMMS Integrated Mathematics*. In S. L. Senk & D. R. Thompson (Eds.), *Standards-based school mathematics curricula: What are they? What do students learn?* (pp. 399–423). Mahwah, NJ: Lawrence Erlbaum Associates.

McCaffrey, D. F., Hamilton, L. S., Stecher, B. M., Klein, S. P., Bugliari, D., & Robyn, A. (2001). Interactions among instructional practices, curriculum, and student achievement: The case of *Standards*-based high school mathematics. *Journal for Research in Mathematics Education*, 32, 493–517.

McLeod, D. B. (1994). Research on affect and mathematics learning in the JRME: 1970 to present. *Journal for Research in Mathematics Education, 34,* 637–647.

Meece, J. L. (1997). *Child and adolescent development for educators*. New York: McGraw-Hill.

National Council of Teachers of Mathematics. *An Agenda for Action: Recommendations for School Mathematics of the 1980s*. Reston, Va: Author.

National Council of Teachers of Mathematics. (1989). *Curriculum and evaluation standards for school mathematics*. Reston, VA: Author.

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

Piaget, J. (1970). Piaget's theory. In P. Mussen (Ed.), *Carmichael's manual of child psychology* (3rd Ed., Vol. 1, pp. 703–732). New York: Wiley.

Raudenbush, S., & Bryk, A. S. (2002). *Hierarchical linear models: Applications and data analysis methods* (2nd Ed.). Thousand Oaks, CA: Sage.

Raudenbush, S. W., Bryk, A. S., & Congdon, R. (2005). *HLM 6.02: Hierarchical linear and nonlinear modeling*. Lincolnwood, IL: Scientific Software International.

Reys, R., Reys, B., Lapan, R., Holliday, G., & Wasman, D. (2003). Assessing the impact of *Standards*-based middle grades mathematics curriculum material on student achievement. *Journal for Research in Mathematics Education, 34*, 74–95.

Riordan, J. E., & Noyce, P. E. (2001). The Impact of two *Standards*-based mathematics curricula on student achievement in Massachusetts. *Journal for Research in Mathematics Education, 32*, 368–398.

Rosenbaum, P. R. (2002). *Observational studies* (2nd ed.). New York: Springer.

Schoen, H. L., & Hirsch, C. R. (2003). The Core-Plus Mathematics Project: Perspectives and student achievement. In S. L. Senk & D. R. Thompson (Eds.), *Standards-based school mathematics curricula: What are they? What do students learn?* (pp. 311–343). Mahwah, NJ: Lawrence Erlbaum Associates.

Schoen, H. L., & Pritchett, J. (1998, April). *Students' perceptions and attitudes in a standards-based high school mathematics curriculum.* Paper presented at the annual meeting of the American Educational Research Association, San Diego, CA. (ERIC Document Reproduction Service No. ED420518).

Schoen, H. L., Cebulla, K. J., Finn, K. F., & Fi, C. (2003). Teacher variables that relate to student achievement when using a standards-based curriculum. *Journal for Research in Mathematics Education, 34*, 228–259.

Schoen, H. L., Fey, J. T., Hirsch, C. R., & Coxford, A. F. (1999). Issues and options in the math wars. *Phi Delta Kappan, 80*, 444–453.

Schoenfeld, A. H. (1989). Explorations of students' mathematical beliefs and behavior. *Journal for Research in Mathematics Education, 20*, 338–355.

Senk, S. L., & Thompson, D. R. (Eds.). (2003). *Standards-based school mathematics curricula? What are they? What do students learn?*. Mahwah, NJ: Lawrence Erlbaum Associates.

Sirin, S. R. (2005). Socioeconomic status and academic achievement: A meta-analytic review of research. *Review of Educational Research, 75*, 417–453.

Stanford Technical Data Report. (1997). Stanford achievement test series (ninth edition). San Antonio, TX: Harcourt Brace.

Tate, W. F. (1997). Race-ethnicity, SES, gender, and language proficiency trends in mathematics achievement: An update. *Journal for Research in Mathematics Education, 28*, 652–679.

Truitt, B. A. D. (1998). *How teachers implement the instructional model in a reformed high school mathematics classroom.* Unpublished doctoral dissertation, University of Iowa, Iowa City.

Tudge, J. R. H., & Winterhoff, P. A. (1993). Vygotsky, Piaget, and Bandura: Perspectives on the relations between the social world and cognitive development. *Human Development*, 36, 61–81.

Webb, N. L (2003). The impact of the *Interactive Mathematics Program* on student learning. In S. L. Senk & D. R. Thompson (Eds.), *Standards-based school mathematics curricula: What are they? What do students learn?* (pp. 375–398). Mahwah, NJ: Lawrence Erlbaum Associates.

Wiley D. E., & Resnick, L. B. (1998) *The new standards reference examination standards-referenced scoring system.* Los Angeles: National Center for Research on Evaluation, Standards, and Student Testing, Graduate School of Education & Information Studies, University of California, Los Angeles.

Wilson, M., & Lloyd, G. (1995). *High school teachers' experiences in a student-centered mathematics curriculum.* Proceedings of the Annual Meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education. (ERIC Document Reproduction Service No. ED389603)

Wu, H. (1997). The mathematics education reform: Why you should be concerned and what you can do. *American Mathematical Monthly, 104,* 946–954.

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

Authors

Michael R. Harwell, 323 Burton Hall, University of Minnesota, Minneapolis, MN 55455; harwe001@umn.edu

Thomas R. Post, 175 Peik Hall, University of Minnesota, Minneapolis, MN 55455; postx001@umn.edu

Yukiko Maeda, 330 Burton Hall, University of Minnesota, Minneapolis, MN 55455; maed0009@umn.edu

Jon D. Davis, 4422 Everett Tower, Western Michigan University, Kalamazoo, MI 49008; jon.davis@wmich.edu

Arnold L. Cutler, 175 Peik Hall, University of Minnesota, Minneapolis, MN 55455; cutler@umn.edu

Edwin Andersen, 175 Peik Hall, University of Minnesota, Minneapolis, MN 55455; ander630@umn.edu

Jeremy A. Kahan, Ida Crown Jewish Academy, 2828 West Pratt Boulevard, Chicago, IL 60645; jkahan@icja.org

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

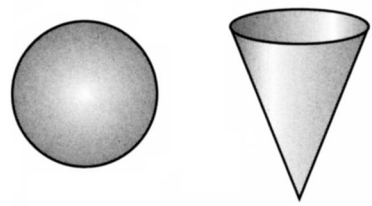
APPENDIX A

Sample Test Items from the New Standards Reference Examination in Mathematics

High School Short Task—Ice Cream Cone

In this task you are asked to explore if the melted ice cream will fit into the cone or if it will spill over.

The ice cream and cone below are drawn accurately and at full size. If the scoop of ice cream is placed on top of the cone and it melts down into the cone, will all the ice cream fit inside the cone? Show whether the melted ice cream will fit inside the cone. For full credit, you must show your calculations and reasoning.



High School Long Task—Shoelaces

You work in a factory that makes shoes. You have to decide what length of shoelace to select when you know the number of lace holes in a shoe. The shoes can have up to ten pairs of lace holes.

1. Estimate the length of shoelace you need for the shoe shown on the next page. (It is shown full size.) Remember to allow enough extra length for tying a bow.
2. Create a rule that will tell you the length of shoelace you need when the number of pairs of lace holes in a shoe is given. Explain how you arrived at your rule.

[A full-page image was used in problem solving.]



3. Now design three signs that can be displayed in the factory to help people select the correct length of shoelaces if they know the number of pairs of lace holes.

   Your sign should describe your rule as clearly and simply as possible.

   (a) One sign must use a table.

   (b) One sign must use a graph.

   (c) One sign must use a formula.
4. Which of these signs do you think would be the most useful in the factory? Give reasons for your answer.

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions

APPENDIX B

Sample Test Items from the SAT-9

SAT- 9 High School Edited Sample items

1. When Mr. Tillen meets a client, the probability that he will make a sale is 1/4. How many sales can he expect if he meets 144 clients?

   (A) 145 (B) 36 (C) 24 (D) 12

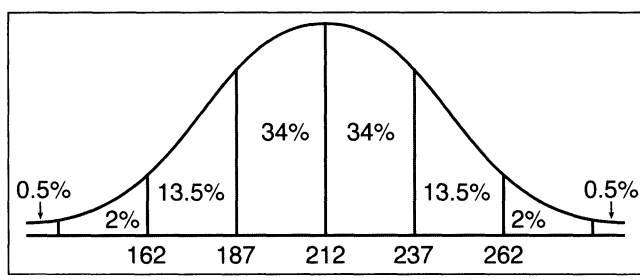
2. After t seconds, the velocity, v, of a basketball thrown upward at 256 feet per second is given by the equation

   $$v = 128 - 32t.$$

   After how many seconds will the velocity equal zero?

   (F) 4 (G) 6 (H) 8 (J) 12

3. The amount of water consumed by single-family homes is distributed as shown.



Water Consumption in Thousands of Gallons per Year

   What is the probability that a single-family home will use *more* than 187 thousand gallons per year?

   (A) 97% (B) 84% (C) 50% (D) 16%

This content downloaded from 134.84.183.174 on Mon, 7 Apr 2014 15:38:42 PM
All use subject to JSTOR Terms and Conditions